BEAGLEBONE

U1
BEAGLEBONEBLACK

VDD_3V3B P9_3 — 3V3, 5V
VDD_5V P9_5 — PWR_FLAG
SYS_5V P9_7
VDD_ADC P9_32 — PWR_FLAG
PWR_BUT P9_9
SYSRST P9_10 — SYS_RESET
AIN0 P9_39 — Temp
AIN1 P9_40
AIN2 P9_37
AIN3 P9_38
AIN4 P9_33 — I2C_SCL
AIN5 P9_36
AIN6 P9_35
B_UART0_RX J1_4
B_UART0_TX J1_5
I2C1_SCL P9_17 — R1 3k — 3V3
I2C1_SDA P9_18 — R2 3k — I2C_SDA
I2C2_SCL P9_19
I2C2_SDA P9_20
SPI1_CS0 P9_28
SPI1_D0 P9_29
SPI1_D1 P9_30
SPI1_SCLK P9_31
EHRPWM1A P9_14
EHRPWM1B P9_16
EHRPWM2A P8_19
EHRPWM2B P8_13
GPIO1_0 P8_25 — CAN_FAULT
GPIO1_1 P8_24 — GPIO1_1
GPIO1_2 P8_5
GPIO1_3 P8_6
GPIO1_4 P8_23 — MRV
GPIO1_5 P8_22 — RTCC_MFP
GPIO1_6 P8_3
GPIO1_7 P8_4
GPIO1_12 P8_12 — EDFA
GPIO1_13 P8_11 — BATT_HEAT_EN
GPIO1_14 P8_16 — GPIO1_14
GPIO1_15 P8_15 — GPIO1_15
GPIO1_16 P9_15 — GPIO1_16
GPIO1_17 P9_23
GPIO1_28 P9_12 — GPIO1_28
GPIO1_29 P8_26 — GPIO1_29
GPIO1_30 P8_21 — GPIO1_30
GPIO1_31 P8_20 — GPIO1_31
UART1_TXD P9_24 — DCAN1RX
UART1_RXD P9_26 — DCAN1TX
UART2_TXD P9_21 — UART_TX
UART2_RXD P9_22 — UART_RX
UART4_TXD P9_13
UART4_RXD P9_11
UART5_TXD P8_37
UART5_RXD P8_38
UART3_CTSN P8_36
UART3_RTSN P8_34
UART4_CTSN P8_35
UART4_RTSN P8_33
UART5_CTSN P8_31
UART5_RTSN P8_32
TIMER4 P8_7
TIMER7 P8_8
TIMER5 P8_9
TIMER6 P8_10
GPIO0_26 P8_14
GPIO0_27 P8_17
GPIO0_7/GPIO3_18 P9_42 — GPIO0_7
GPIO3_19 P9_27
CLKOUT2/GPIO3_20 P9_41 — GPIO3_20
GPIO3_21 P9_25
GPIO2_1 P8_18 — GPIO2_1
GPIO2_6 P8_45
GPIO2_7 P8_46 — 10MhZ_STDBY
GPIO2_8 P8_43 — GPIO2_8
GPIO2_9 P8_44 — GPIO2_9
GPIO2_10 P8_41 — GPIO2_10
GPIO2_11 P8_42 — GPIO2_11
GPIO2_12 P8_39
GPIO2_13 P8_40 — SEL_MODE
GPIO2_22 P8_27 — GPIO2_22
GPIO2_23 P8_29 — NANDWPI
GPIO2_24 P8_28 — GPIO2_24
GPIO2_25 P8_30 — NAND_HOLD
NC J1_2
SHIELD P1 — C11 5nF — 1M — GND
DGND J1_1
GNDA_ADC P9_34 — R3 — GND
GND

PC104 HEADERS

J8
Conn_02x26_Odd_Even

CANL 1	2		
CANH 3	4		
5	6		
7	8		
RS45A 9	10		
RS45B 11	12 — GND		
BATT_HEAT_EN 13	14 SYSRST	BAT_BRD_TMP 53	54 PDU_TMP
GPIO1_26 15	16 GPIO2_8	55	56
GPIO1_15 17	18 GPIO2_11	57	58
UART_RX 19	20 UART_TX	59	60
5V_S 21	22 BATT_FAULT	61	62
5V_S 23	24 CHRG	63	64
SWITCH_MODE_SEL 25	26 SOL_SHDN	65	66
3V3_S_EN 27	28 BURN1_N	67	68
BURN_IN 29	30 BURN2_N	GPIO1_14 69	70 ADCS_BOOT
5V_S_EN 31	32	EDFA_5V_STAT 71	72 ADCS_ENABLE
UART_TX 33	34 GPIO2_9	5V_EDFA_SW 73	74 5V_EDFA_SW
UART_RX 35	36 GPIO1_27	5V_BUS_STAT 75	76 5V_BUS_STAT
3V3 37	38	5VIN 77	78 5VIN
39	40	3V3IN 79	80 3V3IN
I2C_SDA 41	42	81	82
I2C_SCL 43	44	83	84 — GND
GPIO2_24 45	46 SWITCH_SHUTDOWN	85	86
SW_3V3 47	48 GPIO2_1	87	88
SW_3V3 49	50 GPIO2_22	89	90
SBWTDIO 51	52 SBWTCK	91	92
		93	94
		95	96
		97	98
		99	100
		101	102
		103	104

GND — GND

CAN TRANSCEIVER

C3 100nF — C1 1uF — GND — 3V3
C2 10uF
U3
TCAN337G
DCAN1TX — R4 62 — TXD
DCAN1RX — R5 62 — RXD
CAN_FAULT — FAULT
VCC
GND
CANH — R6 62 — C5 10nF — GND
CANL — R7 62

RTCC W BATTERY BACKUP

3V3
C6 100nF
PWR_FLAG
BT1 3V3
GND
I2C_SCL — SCL
I2C_SDA — SDA
RTCC_MFP — MFP
VCC
VBAT
VSS
X1
X2
TP2 TestPoint
XTAL1
SIT1533AI-H4-DCC-32.768E
U4
MCP7940N-xSN
NC
GND
Vdd
CLK_Out
3V3
GND

OVERVOLTAGE PROTECTION

3V3IN
Q1
AON6411
PWR_FLAG
TestPoint
3V3
U2
TPS2400DBVR
TP3 TestPoint
VIN
GATE
GND
GND
5VIN
Q2
AON6411
PWR_FLAG
TestPoint
3V3
U8
TPS2400DBVR
TP5 TestPoint
VIN
GATE
GND
GND

RS485 BACKUP CAN

U5
ADM3485EARZ
3V3
UART_RX — RO
GPIO1_1 — RE*
DE
DI
UART_TX
VCC
B
A
GND
R11 10k — GND
RS45A — R9 62 — C4 10nF — GND
RS45B — R8 62

MOUNTING HOLES

H1 MountingHole_Pad
H2 MountingHole_Pad
H3 MountingHole_Pad
H4 MountingHole_Pad
GND

WATCHDOG TIMER (Not Processor)

U6
MAX706RESA+T
TP1 TestPoint
3V3
MRV — *MR
VCC
GND
PFI
*WDO — SYSRST
*RESET — GPIO3_20
WDI
*PFO
GND

TEMP SENSOR

C10 100nF
R10 62
3V3
U7
TMP36xS
+Vs
SHUTDOWN
VOUT — Temp
GND
GND

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